

Statement of Purpose

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When a smart-traffic agency reports a 16–37 percent improvement in corridor flow, the unresolved economic question is not whether the system changed operations, but whether the change can be attributed to the intervention after accounting for rollout timing, untreated controls, spillovers, pre-existing trends, and cost valuation. A traffic-performance claim is not automatically a welfare gain. It becomes policy evidence only when the counterfactual is credible and when monetized benefits are compared with capital cost, operating cost, time savings, reliability, safety, emissions, and uncertainty. This distinction is the reason I am applying to the Master of Science in Economics and Policy at the American University of Sharjah. My goal is to develop the econometric and policy-evaluation training needed to assess smart-mobility investments in Dubai and similar urban settings with stronger identification discipline than my previous applied work could provide.

My first research trajectory began with a final-year project at the Université Internationale de Casablanca on the financial profitability of outdoor advertising assets in Casablanca. The practical problem was that advertising operators lacked a structured method for deciding where to invest in static OOH or digital DOOH billboards under traffic, demographic, regulatory, and cost constraints. My hypothesis was that high-traffic corridors and high-purchasing-power zones would generate higher expected returns, but only if the visibility premium survived CAPEX, OPEX, taxes, maintenance, energy use, and compliance requirements. I built a mixed-methods appraisal model using SUMO traffic simulation, GIS mapping, demographic and economic proxies, and five-year financial modeling, then constructed a Potential Profitability Index to compare locations. The results showed that high-congestion and affluent districts offered stronger investment potential, while DOOH assets were attractive only when higher upfront costs were offset by content flexibility, slot monetization, and regulatory compliance. This project gave me my first exposure to spatial investment appraisal, but it also exposed the boundary of ROI-based decision models: they can rank profitable locations, yet they cannot establish whether an intervention changed welfare relative to a counterfactual. The outcome was a completed thesis and a decision model that turned traffic, spatial, and financial variables into an investment-prioritization tool.

That limitation shaped the next stage of my work. In an independent paper on AI-MCDM investment scoring for smart-city infrastructure under carbon, energy, and geoeconomic risk, I moved from location profitability toward broader infrastructure prioritization. The research question was whether smart-city projects could be ranked more transparently when machine-learning forecasts, carbon-price scenarios, real-options logic, and multi-criteria decision methods were integrated into a single auditable scorecard. My hypothesis was that XGBoost would improve forecasting relative to linear baselines, that carbon-price scenarios would alter project rankings, and that geoeconomic-risk proxies would reduce risk-adjusted investment scores in exposed settings. I designed a four-layer framework combining XGBoost-based demand forecasting, carbon-adjusted Decoupled Net Present Value, Real Options valuation, Fuzzy-AHP/MEREC weighting, and AROMAN ranking, then benchmarked the project sample against World Bank, EIB, WDI, and ESG data. XGBoost reduced mean absolute error by 45.8 percent relative to Ridge regression in the original temporal validation, and the AROMAN ranking diverged from NPV-only ordering across all 25 projects. Yet the most important result was not the performance gain. Early AROMAN implementations produced unstable rankings, and the geoeconomic-risk hypothesis was not confirmed at the full-panel level; I corrected the normalization logic, separated prediction from ranking and causal interpretation, and learned to treat negative evidence as methodological information rather than as a result to hide.

This intellectual progression now leads to my proposed research at AUS: a data-gated quasi-experimental evaluation of smart-traffic investments in Dubai. I want to study whether exposure to named systems such as DITSC or UTC-UX is associated with lower delay, lower speed loss, or more reliable travel times, and whether any estimated gains survive cost-benefit appraisal. The project would begin with a treatment registry recording activation timing, geography, exposure level, and source confidence. If treatment timing, corridor geography, outcomes, and controls are strong enough, I would estimate effects using difference-in-differences and event-study models, with pre-trend, matching-balance, contamination, and minimum-detectable-effect diagnostics. If the traffic panel cannot support causal claims, I would step down to secondary property-capitalization evidence using Dubai

Land Department transactions around treated buffers. If that evidence is also too weak, I would report only scenario-based CBA, clearly labeling effects as assumed rather than observed. The objective is not to force a positive finding; it is to build a defensible evidence hierarchy for smart-mobility policy under restricted urban data.

My methodological gap is clear. I have worked with financial modeling, QGIS, SUMO, Excel automation, Power Query, MCDA, and introductory machine-learning workflows. What I now need is formal graduate training in applied econometrics, causal identification, and policy evaluation so that my future work can move from investment scoring to defensible welfare inference. Applied Econometrics I and II would directly support the DiD and event-study component of my proposed design. Time Series Analysis and Forecasting would strengthen the modeling of traffic, demand, and real-estate indicators. Economics of Agglomeration, Real Estate and Urban Development would support the hedonic fallback using Dubai property transactions. Economics of Sustainability and Resource Use would strengthen the welfare-accounting side of the project, especially where emissions, energy use, and public-resource allocation enter the appraisal.

The faculty fit at AUS is unusually relevant to this direction. Professor Mohammad Arzaghi's work in urban economics, real estate economics, applied microeconomics, and location decisions aligns directly with my interest in property-capitalization evidence and Dubai real-estate data as a fallback when traffic microdata are unavailable. His connection to the Dubai Land Department Economic Council makes his profile especially relevant to a project that must translate urban data into policy evidence without overstating what property prices can prove. Professor Ismail Genc's teaching and research in forecasting, applied time series analysis, quantitative methods, and applied economics match the demand, traffic, and market-indicator components of my proposed empirical design. Professor Khusrav Gaibulloev's expertise in econometrics, public economics, mathematical economics, and applied policy research would help me strengthen the identification discipline behind the project, especially where weak data require careful claim downgrading rather than rhetorical overreach.

My internships confirmed that empirical work rarely begins with clean data. In debt collection, advertising analytics, and leasing finance, I worked with fragmented operational datasets, inconsistent reporting structures, and managers who needed interpretable outputs rather than technical elegance. These experiences shaped the practical logic behind my proposed data gate: when evidence is incomplete, the correct response is not to exaggerate the model, but to reduce the claim to the level the data can support. That discipline is what I want to bring into graduate research.

Over the next five to ten years, I aim to work at the intersection of urban economics, infrastructure appraisal, smart-mobility policy, and data-driven public decision-making in the Gulf, North Africa, and other fast-growing urban regions. My longer-term objective is to help design evaluation systems that allow governments, investors, and urban authorities to distinguish operational improvement from welfare gain, simulation from causal evidence, and technology narratives from investments that pass economic appraisal. AUS MSEP is the right environment for this transition. I would enter with applied experience in traffic simulation, spatial analytics, financial modeling, and investment scoring; I would use the program to acquire the economic methodology needed to turn those tools into credible policy evidence.